PROWESS
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USER MANUAL

SPECIAL PURPOSE MACHINE (SPM)

MCU SUB ASSEMBLY STATION



CUSTOMER
Tata Motors Ltd.

MANUFACTURER
Prowess Autotech Pvt. Ltd.

DATE
10-10-2026

READ THIS MANUAL COMPLETELY BEFORE OPERATING THE MACHINE.

SAFE OPERATION | PRECISE PERFORMANCE | PREMIUM QUALITY | EASY MAINTENANCE | RELIABLE RESULTS

PROWESS AUTOTECH PVT. LTD.

LOCATION : RUDRAPUR, UTTARAKHAND
PIN CODE : 263153

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3. OPERATING INSTRUCTIONS

This section describes the operating procedure for **TATA TML Fixture Clamping & Bolt Tightening SPM.**

NOTE
The machine cycle will proceed only when the part is correctly positioned, both push buttons are pressed together, and torque confirmation is accepted on the HMI.

3.1 START-UP PROCEDURE

1. Switch ON the main power supply.
2. Ensure the compressed air supply is ON and pressure is within the approved operating range.
3. Check that the fixture and resting pads are clean and free of obstruction.
4. Confirm the correct component is available for loading.
5. Check that the torque wrench is ready and set to the specified torque.
6. Switch ON the control panel and check the HMI for alarms or faults.
7. Ensure the fixture is in the unclamped/ home condition before loading.

3.2 OPERATING CYCLE

1. Load the component onto the fixture resting pads.
2. Position the component correctly.
3. Press both push buttons simultaneously to clamp.
4. Tighten the bolts using the torque wrench.
5. Verify torque confirmation on the HMI.
6. After accepted confirmation, the fixture automatically de-clamps.
7. Unload the completed part and load the next component.

3.3 OPERATOR GUIDELINES

- Keep hands clear of clamping points and moving parts.
- Press both push buttons together; never bypass the safety function.
- Do not adjust or reposition the component while clamped.
- Use the torque wrench and specified torque value only.
- Confirm torque status on the HMI before unloading.
- Do not force the fixture open or interfere with automatic de-clamping.
- Report abnormal noise, air leakage, or HMI faults.
- Only authorized personnel may perform maintenance.

3.4 EMERGENCY STOP PROCEDURE

1. Press EMERGENCY STOP immediately if an unsafe condition occurs.
2. Keep clear and inform the supervisor/maintenance team.
3. Resolve the fault and reset Emergency Stop only when safe.
4. Restart the cycle only after checks and authorization.

IMPORTANT
Never bypass safety devices. Keep hands clear of the fixture during clamping and de-clamping. Follow site lockout/tagout rules before maintenance.

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6. MAINTENANCE SCHEDULE

Regular maintenance ensures smooth operation, longer machine life and consistent performance of the **TATA TML Fixture Clamping & Bolt Tightening SPM**.

NOTE

- Follow the maintenance intervals mentioned below.
- Use recommended tools and genuine spares where required.
- If any abnormal noise, vibration or fault is observed, stop the machine and inform the supervisor immediately.

INTERVAL	TASK / CHECK POINT	DETAILS	RESPONSIBLE
DAILY (Before Start)	• Clean machine surface. • Check air pressure. • Check push buttons (both hands). • Check HMI display. • Visual check of fixture & resting pads. • Check for air leaks / unusual noise.	Remove dust, chips and debris. Ensure 5 – 6 Bar pressure. Both buttons should work together and not stick. HMI should power ON and display normally. Pads should be clean and properly seated. Listen for abnormal sound / air leak.	Operator
WEEKLY (Every 7 Days)	• Lubricate moving parts (guide rails, etc.). • Check air filter / regulator. • Check pneumatic cylinder operation. • Inspect cables & connectors. • Check torque wrench calibration (visual / functional).	Use recommended lubricant (light machine oil). Clean filter bowl and drain water. Smooth operation, no air leakage. Ensure wiring is tight and undamaged. Verify wrench works properly and gives correct torque (as per set value).	Maintenance (Mechanical + Electrical)
MONTHLY	• Check solenoid valves (operation). • Inspect hoses and fittings. • Check fixture alignment & clamping pads. • Tighten loose bolts and fasteners. • Check electrical panel interior (dust).	Ensure proper response and no leakage. No cracks / wear, proper routing. Fixture should sit correctly, pads undamaged. Check all hardware is tight. Remove dust (if required).	Maintenance (Mechanical + Electrical)
3 MONTHS	• Replace air filter element (if required). • Check pneumatic seals / cylinder rod. • Inspect wiring, relays and contactors. • Verify HMI touch response.	Use genuine filter element. Check for wear, leakage or damage. Clean and tighten terminals. Check display, touch and buttons.	Maintenance (Mechanical + Electrical)
6 MONTHS	• Check complete pneumatic system (pressure, leaks, hoses). • Inspect clamping cylinders for wear. • Check push button safety function. • Check torque wrench accuracy.	Ensure stable pressure and no leakage. Look for seal wear or smooth movement. Both buttons and safety circuit must work properly. Calibrate / verify torque value (if required).	Maintenance (Mechanical + Electrical)
YEARLY	• Full machine inspection & cleaning. • Replace worn-out parts (as required). • Check electrical panel components. • Verify safety interlocks and emergency stop.	Remove dust and tighten all fittings. Replace damaged parts (pads, seals, switches, etc.). Check wiring, terminals, and components. Test E-Stop and safety function.	Maintenance (Authorized / Qualified)

USE GENUINE SPARES
For better performance and longer life.

KEEP IT CLEAN
Dust and chips reduce accuracy and life.

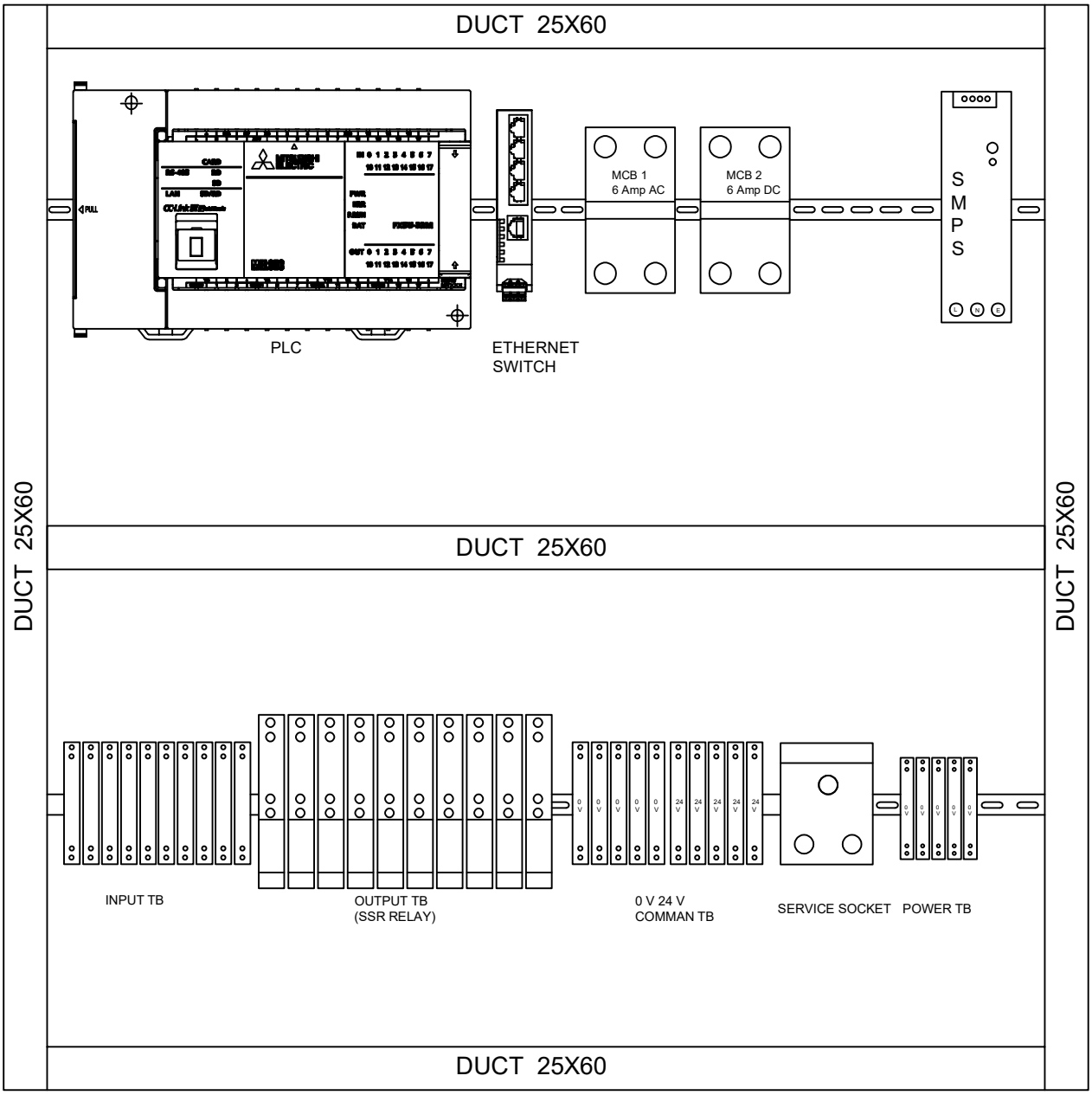
DO NOT BYPASS SAFETY INTERLOCKS
They are for your safety.

FOLLOW TORQUE VALUES
For reliable clamping and quality output.

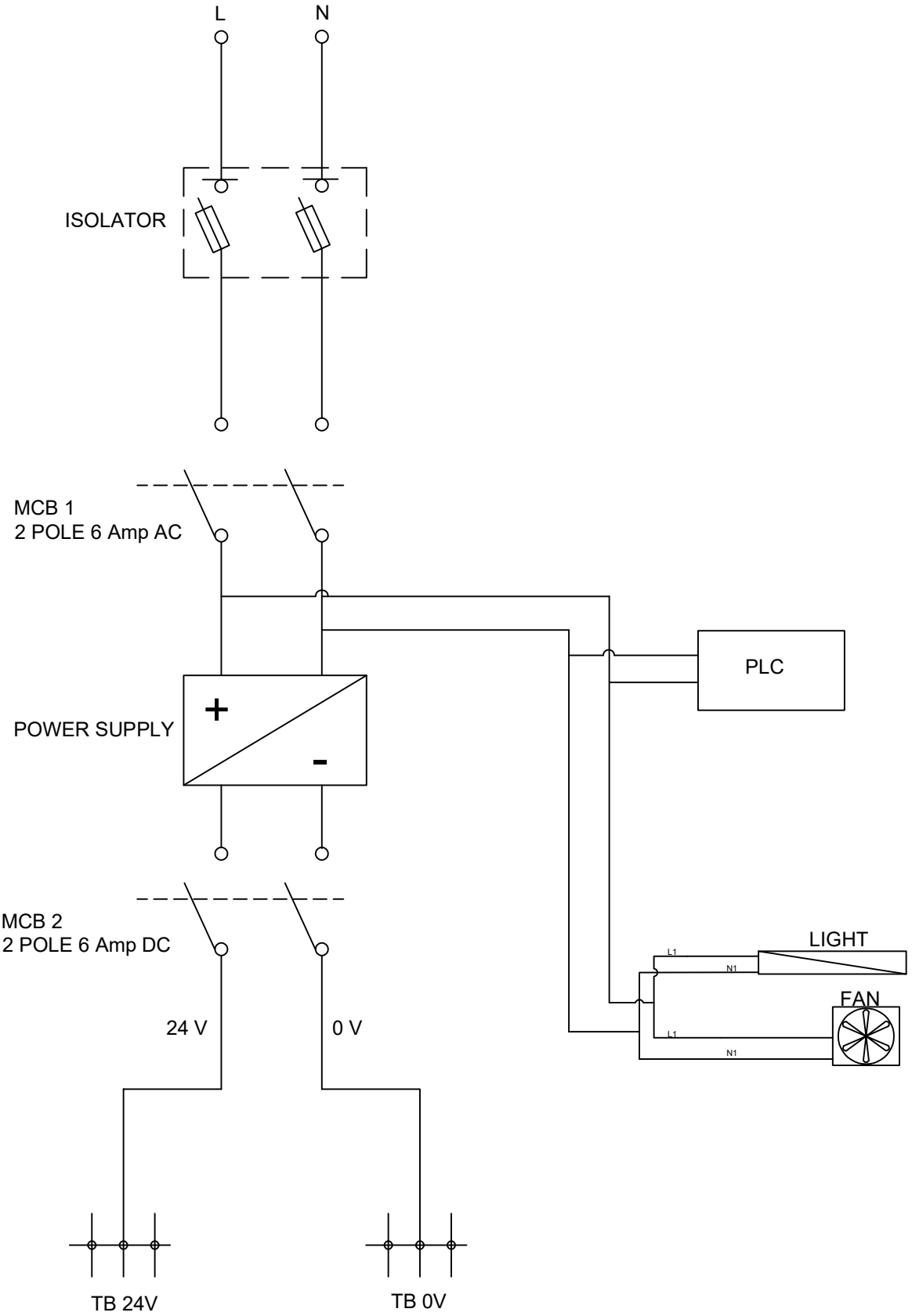
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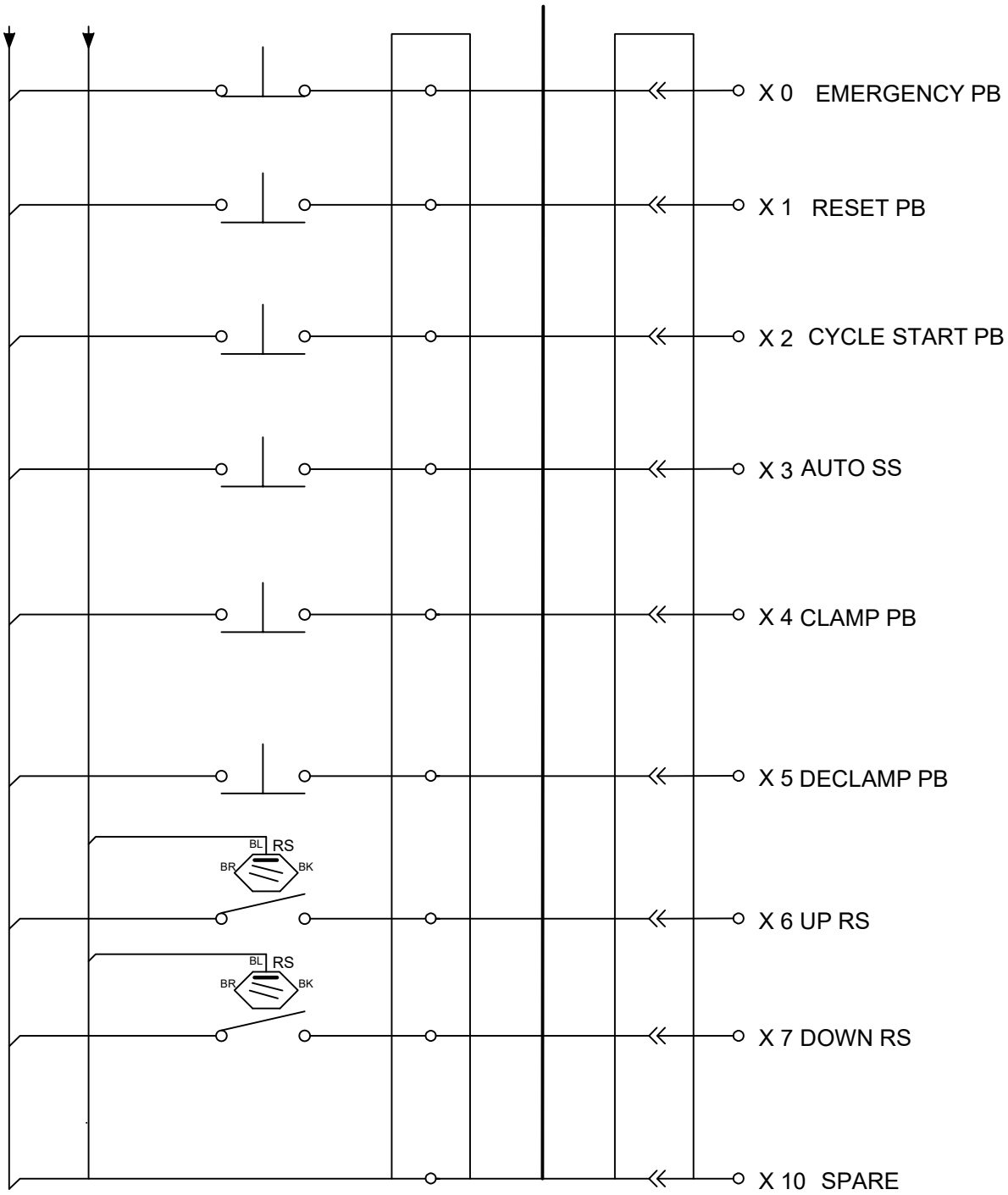


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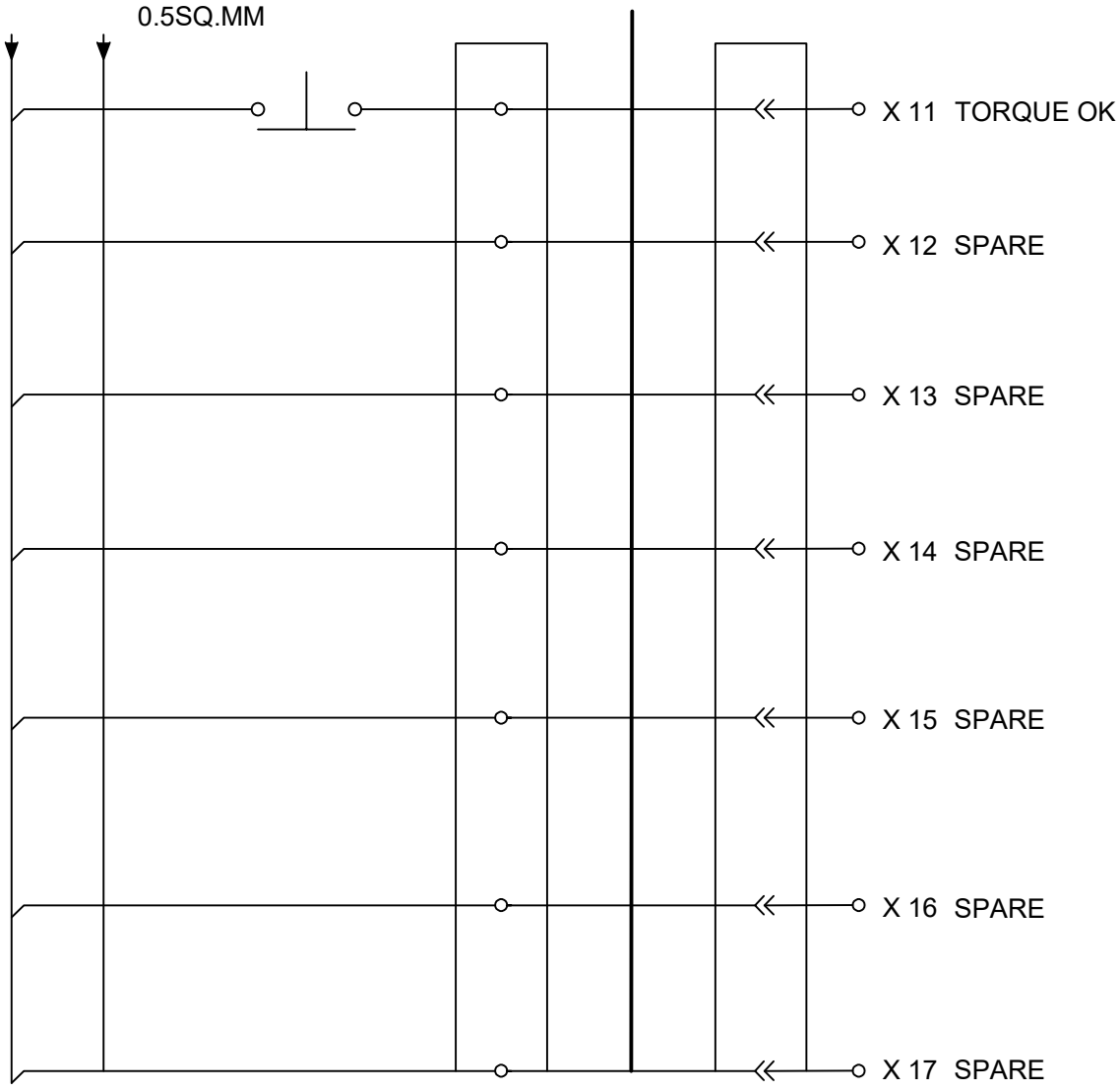


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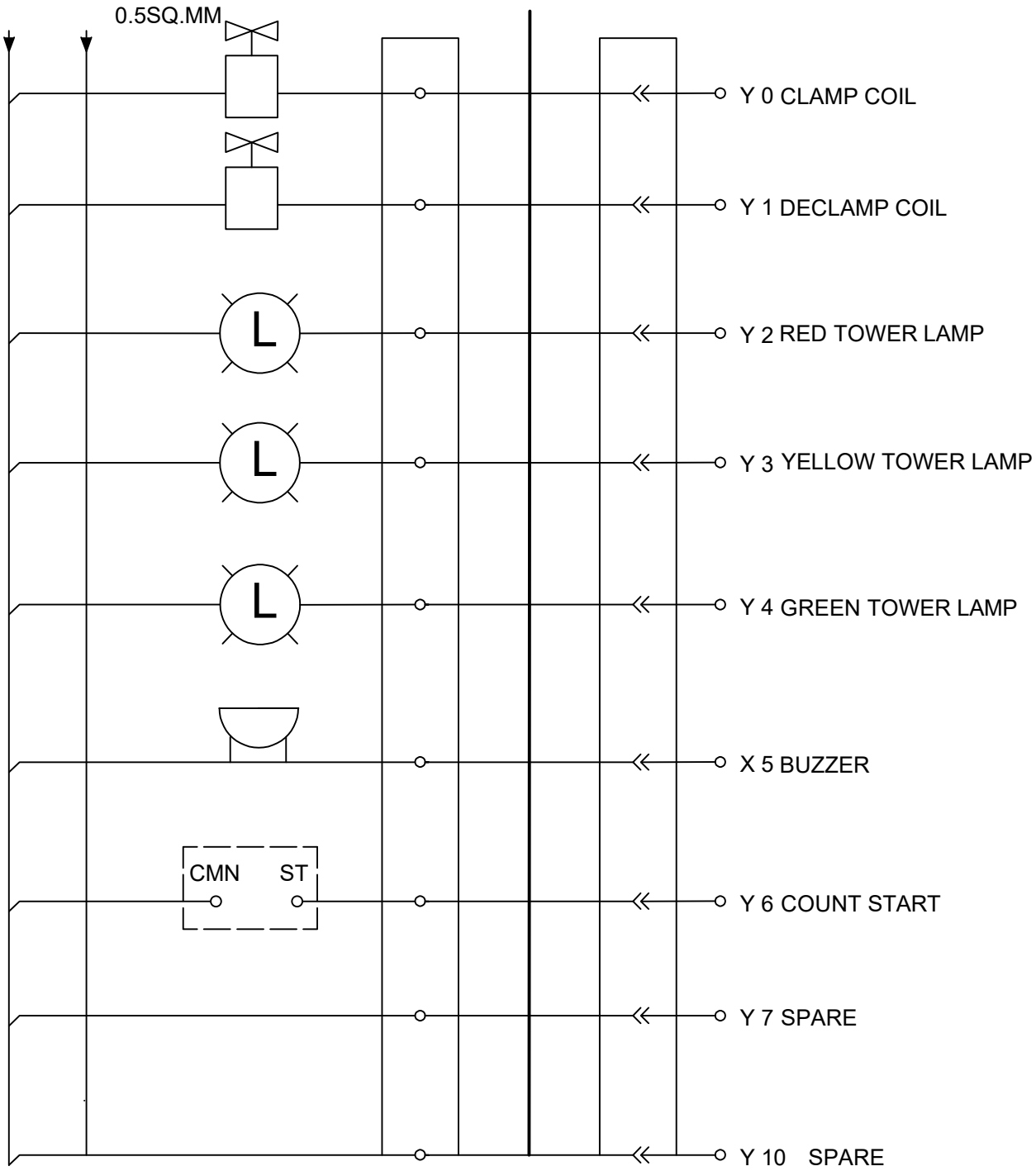
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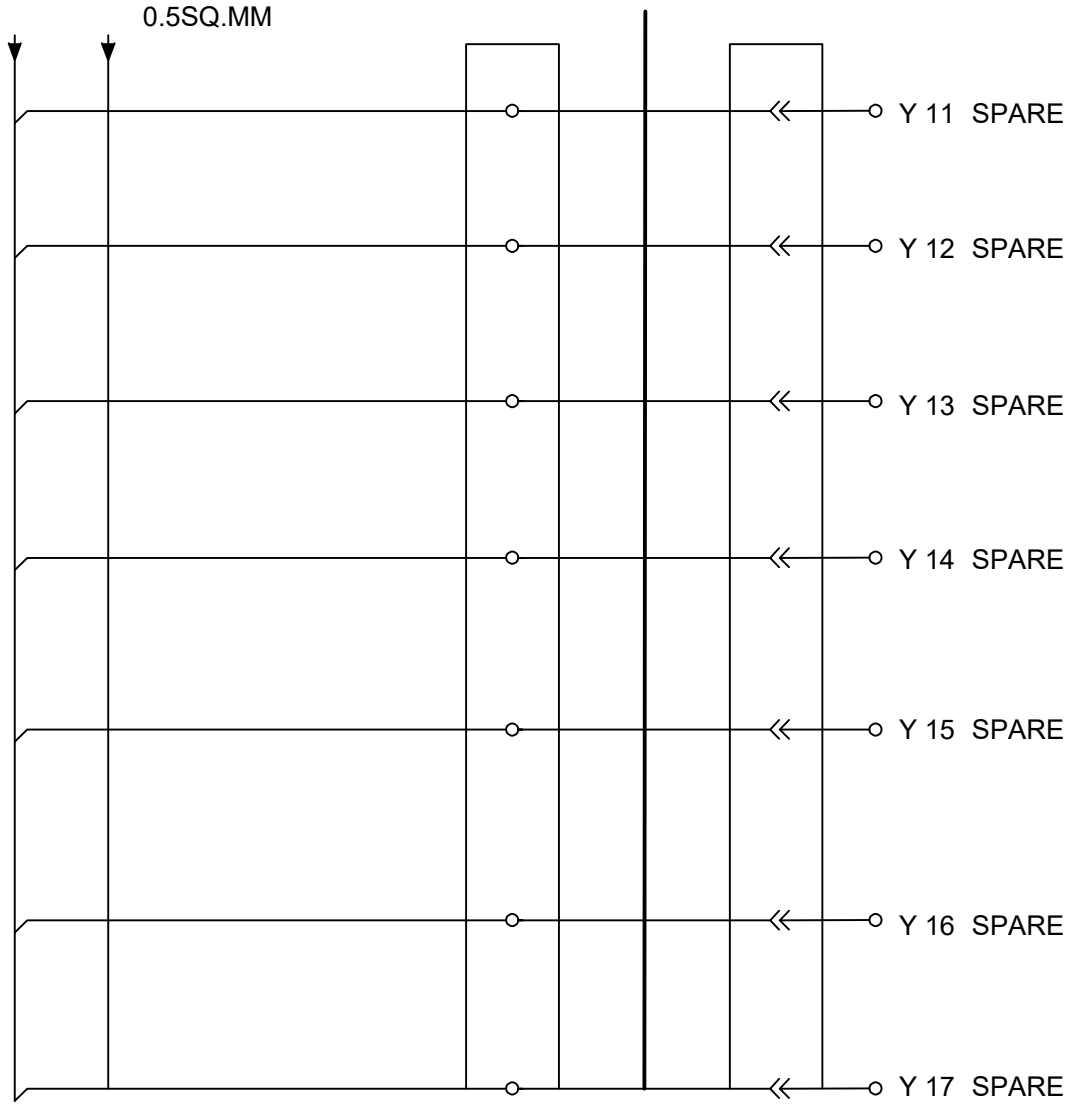
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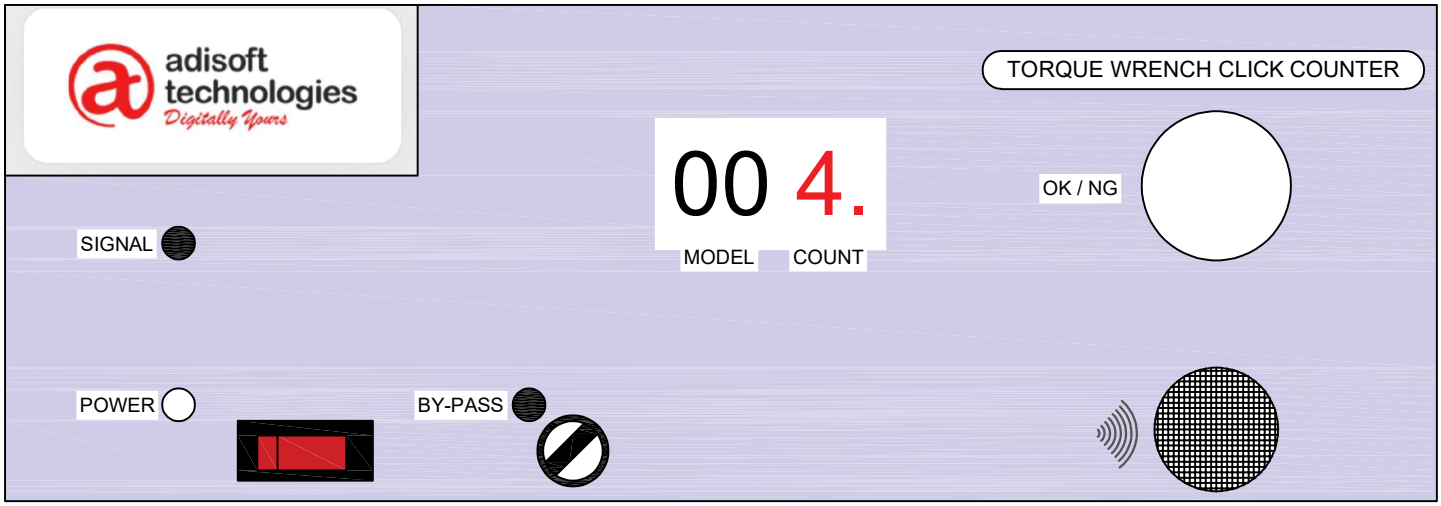
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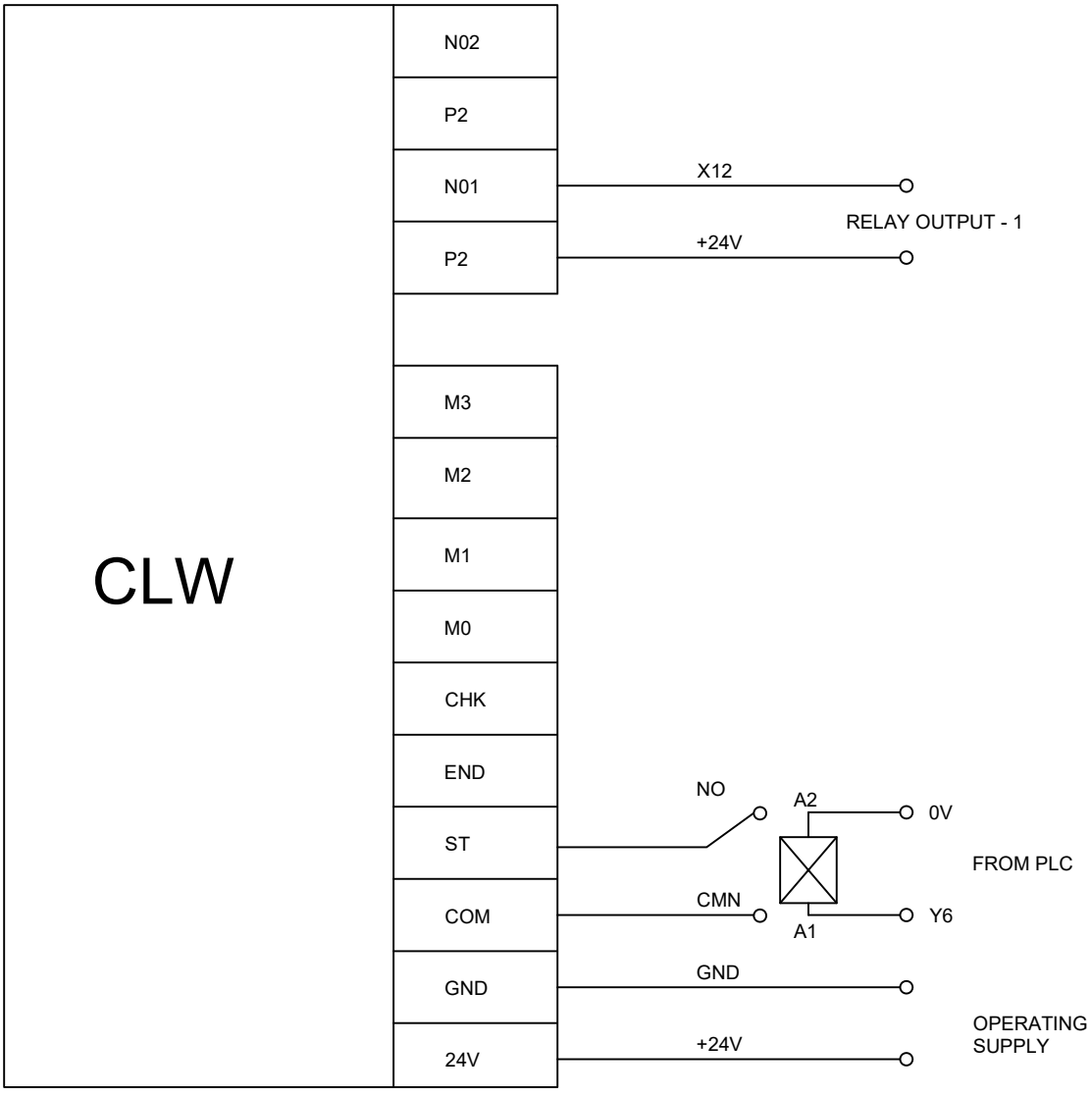
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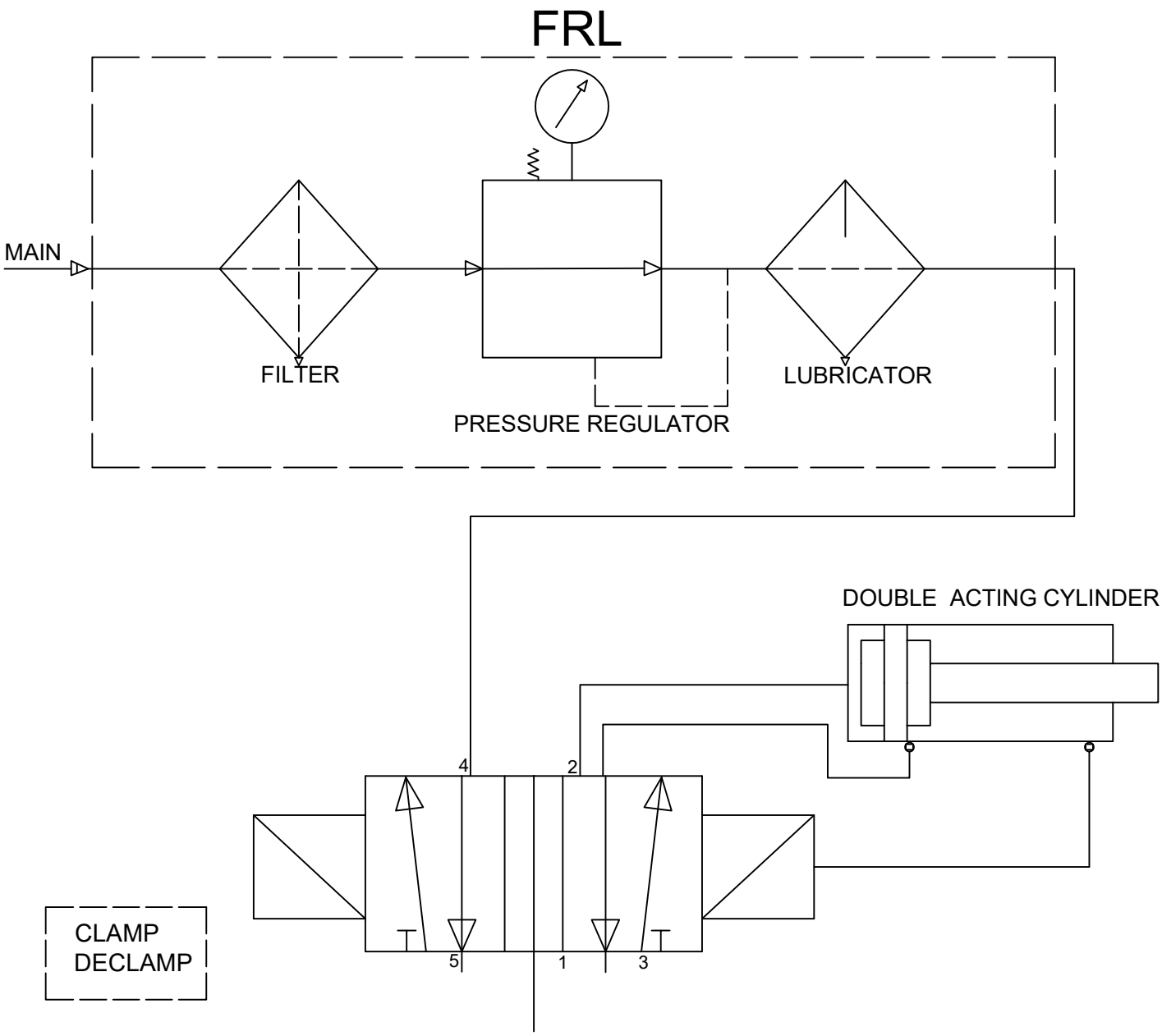
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EXTERNAL CONNECTION



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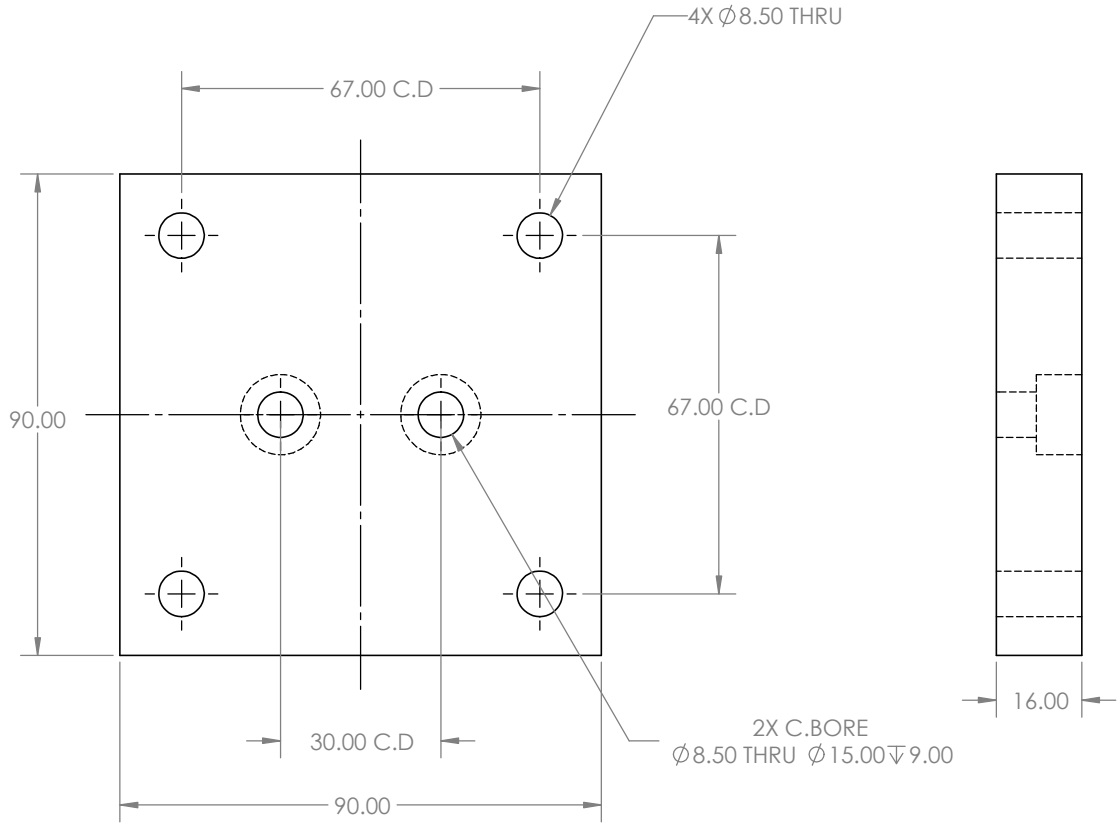
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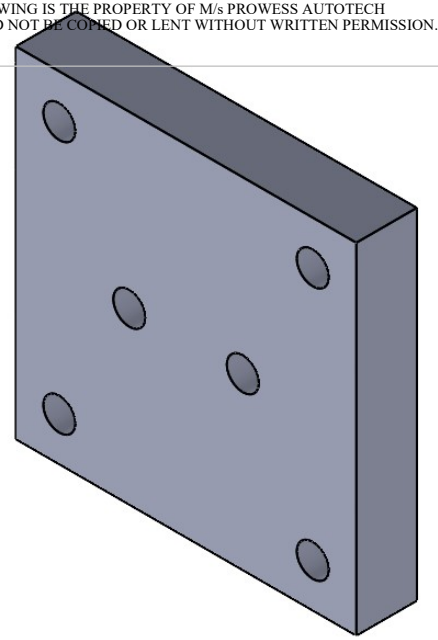
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PAGE : 12

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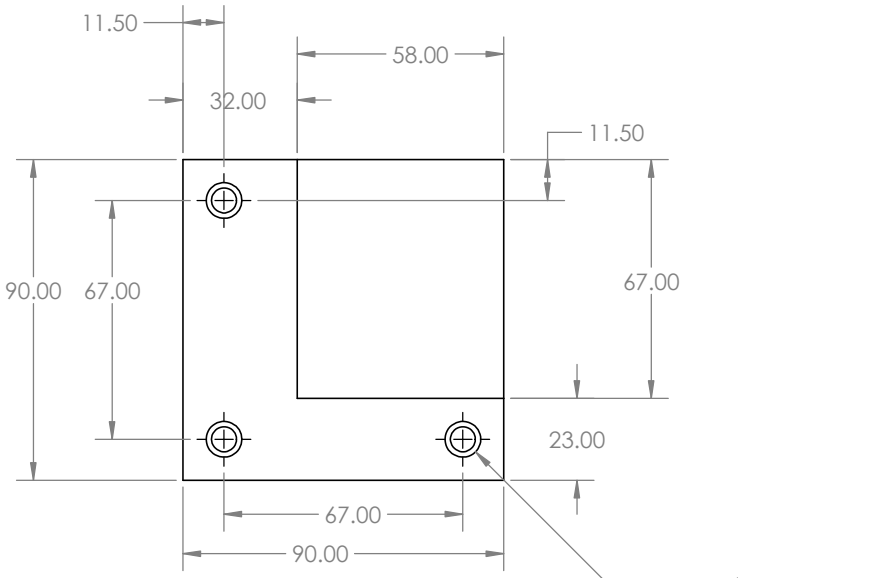
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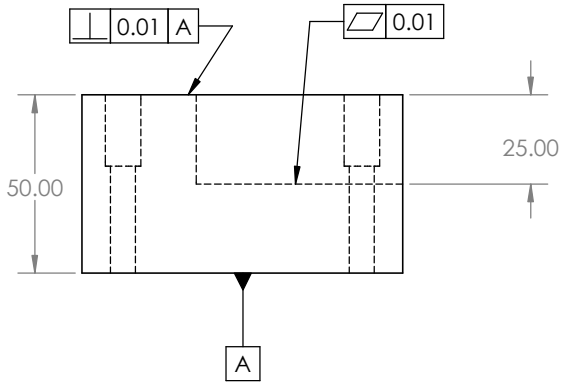
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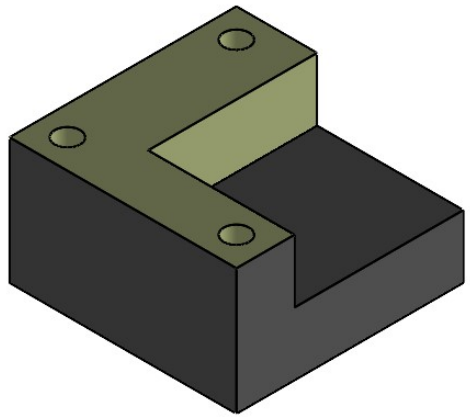
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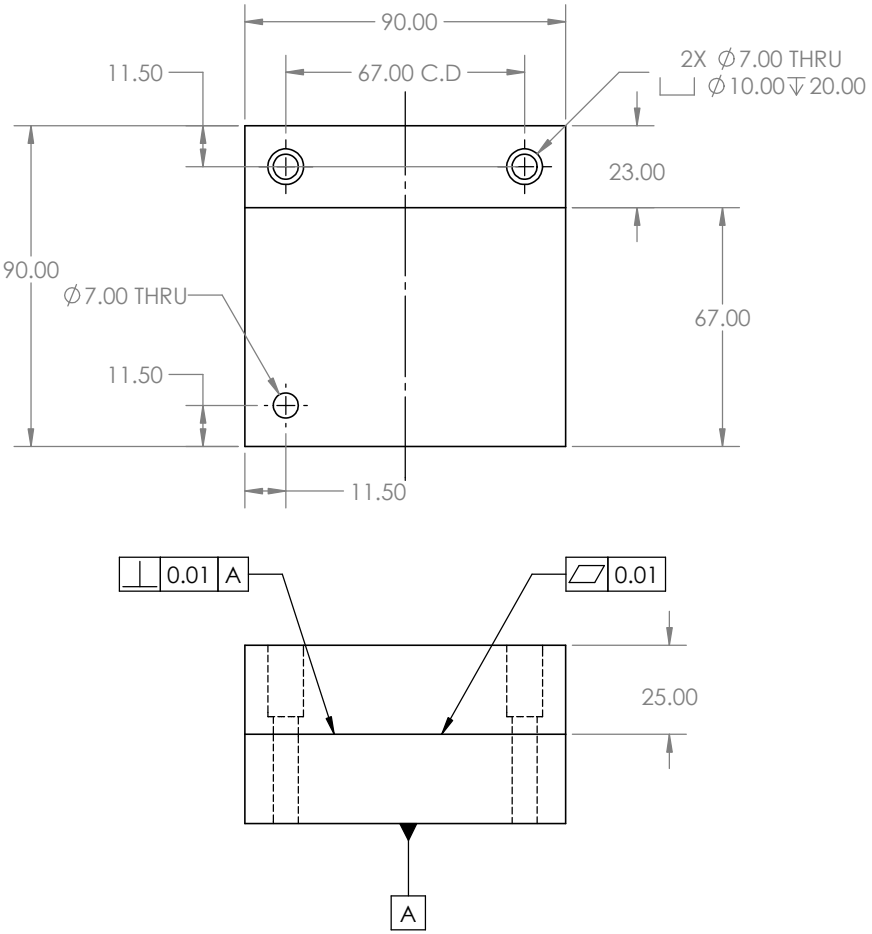
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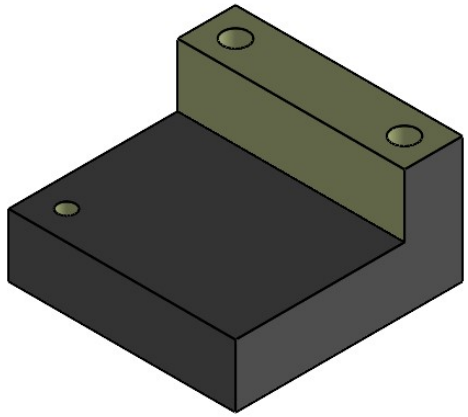
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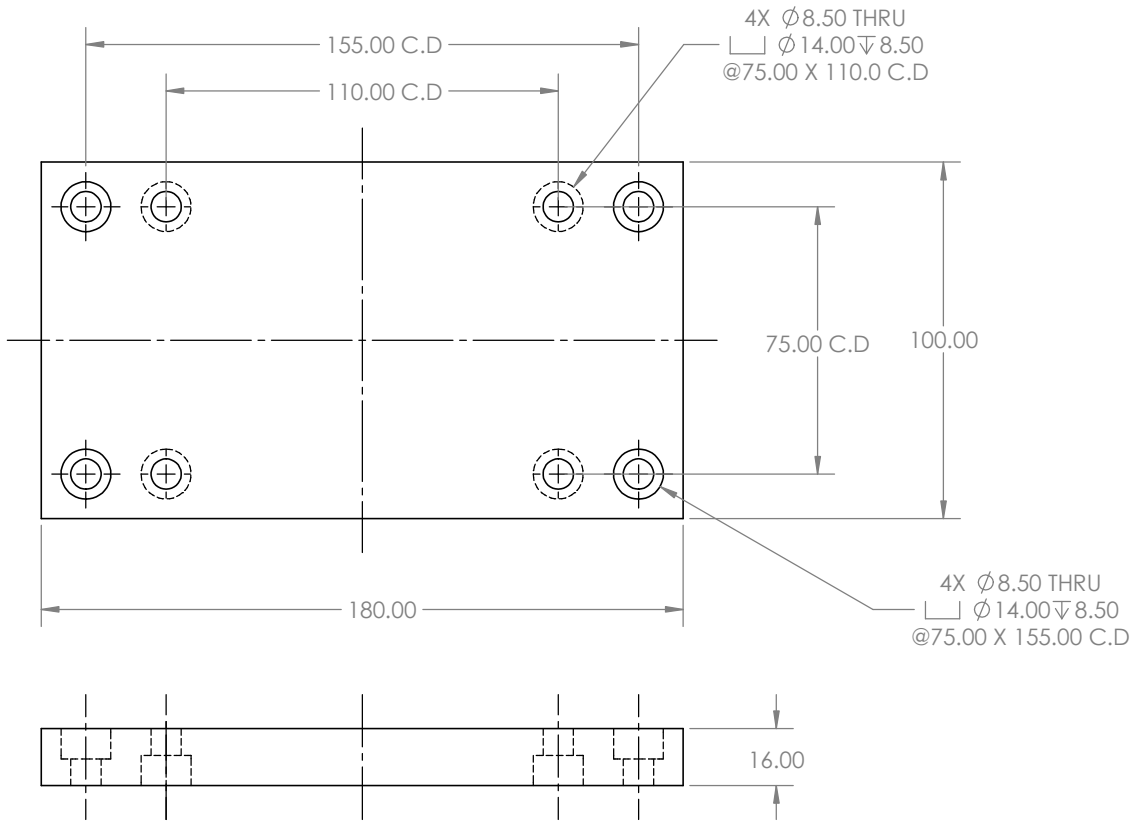
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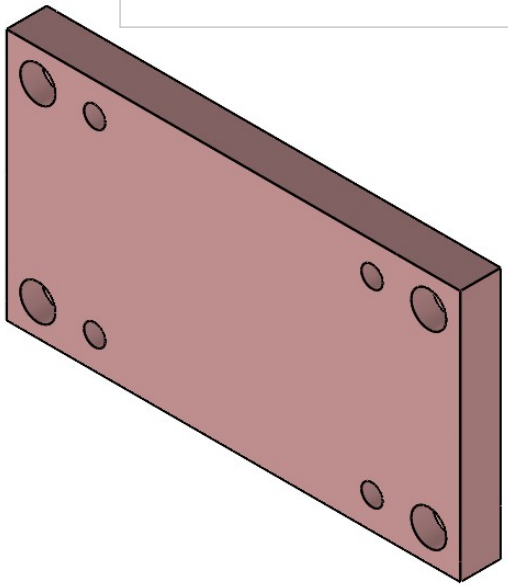
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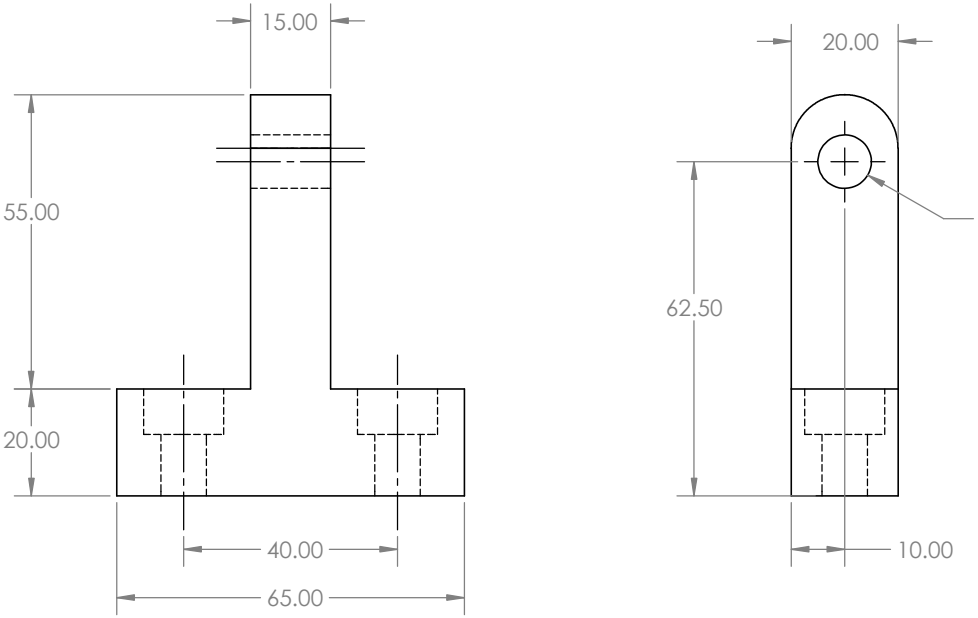
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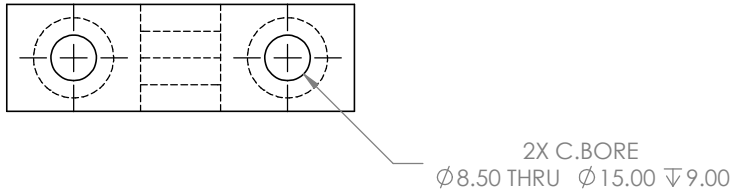
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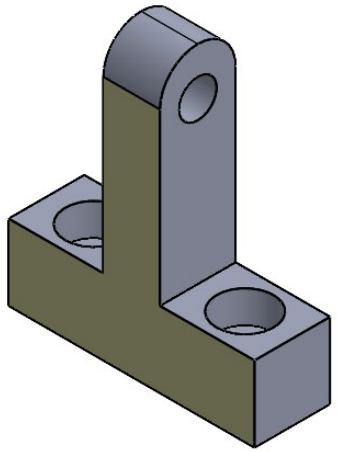
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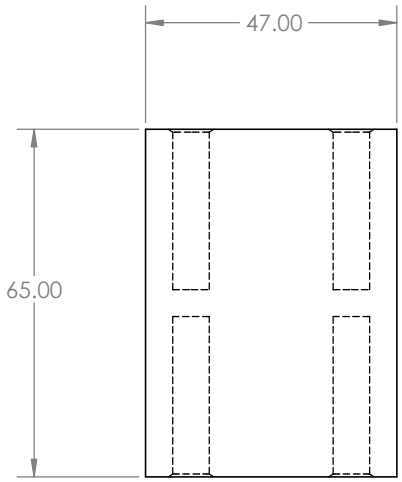
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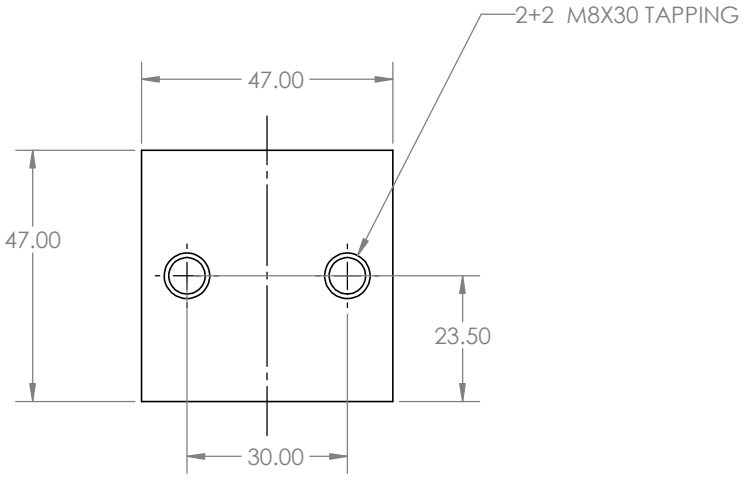
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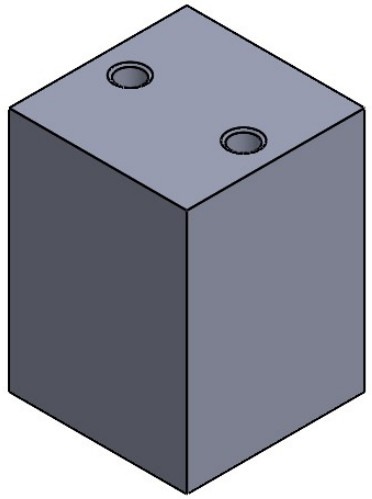
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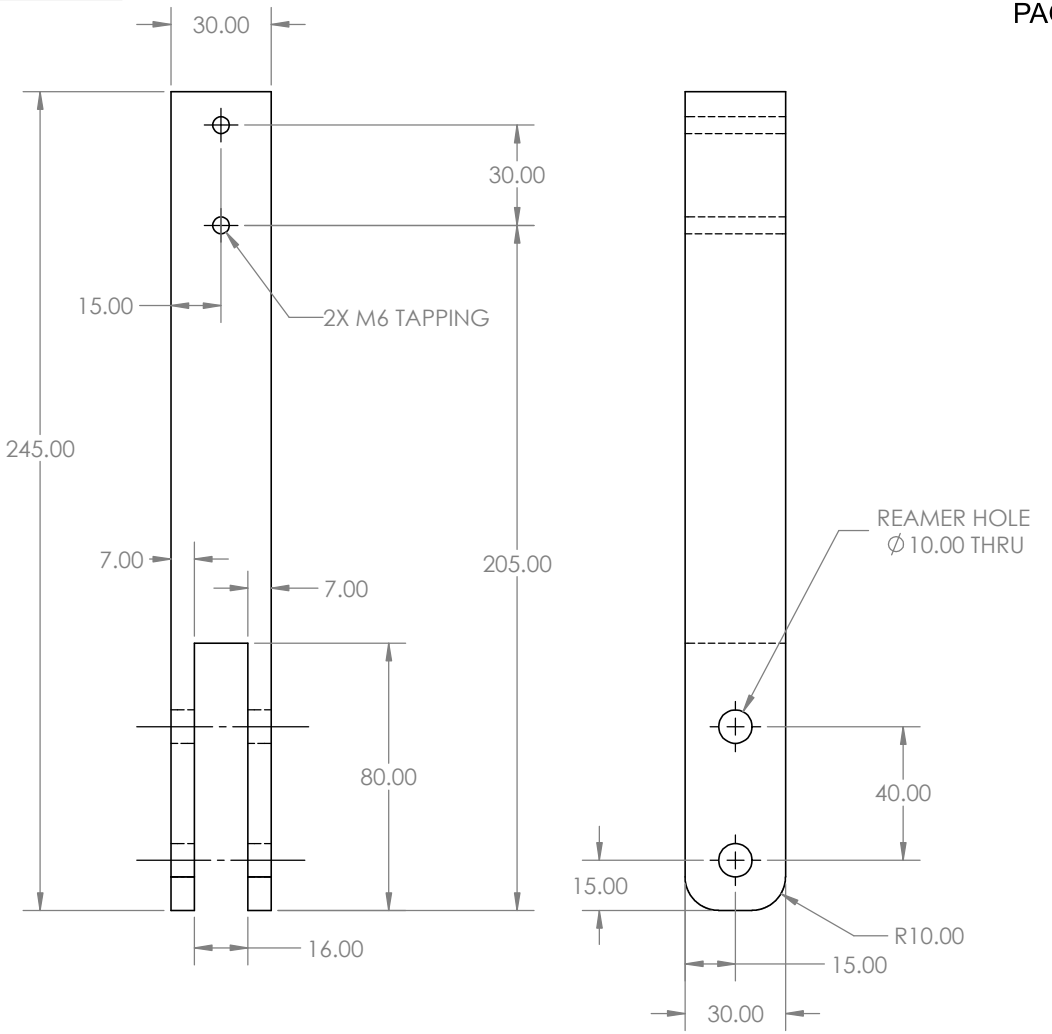
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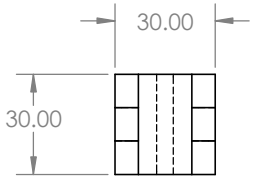
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

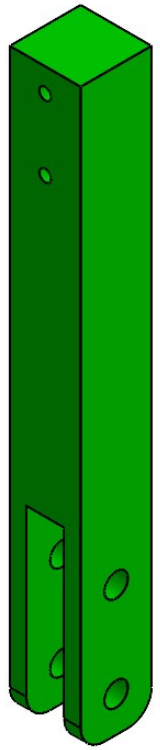
PROWESS AUTOTECH PVT. LTD. OUTSOURCE YOUR PAIN

DO NOT SCALE THE DRG.
ASK IF IN DOUBT.

A
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MATERIAL = BRIGHT BAR
QTY = 1 NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE MCU SUB ATATION ASSEM		Ref Drawing No.	
		Drawing No.	

Rev

Prowess Autotech Pvt. Ltd.

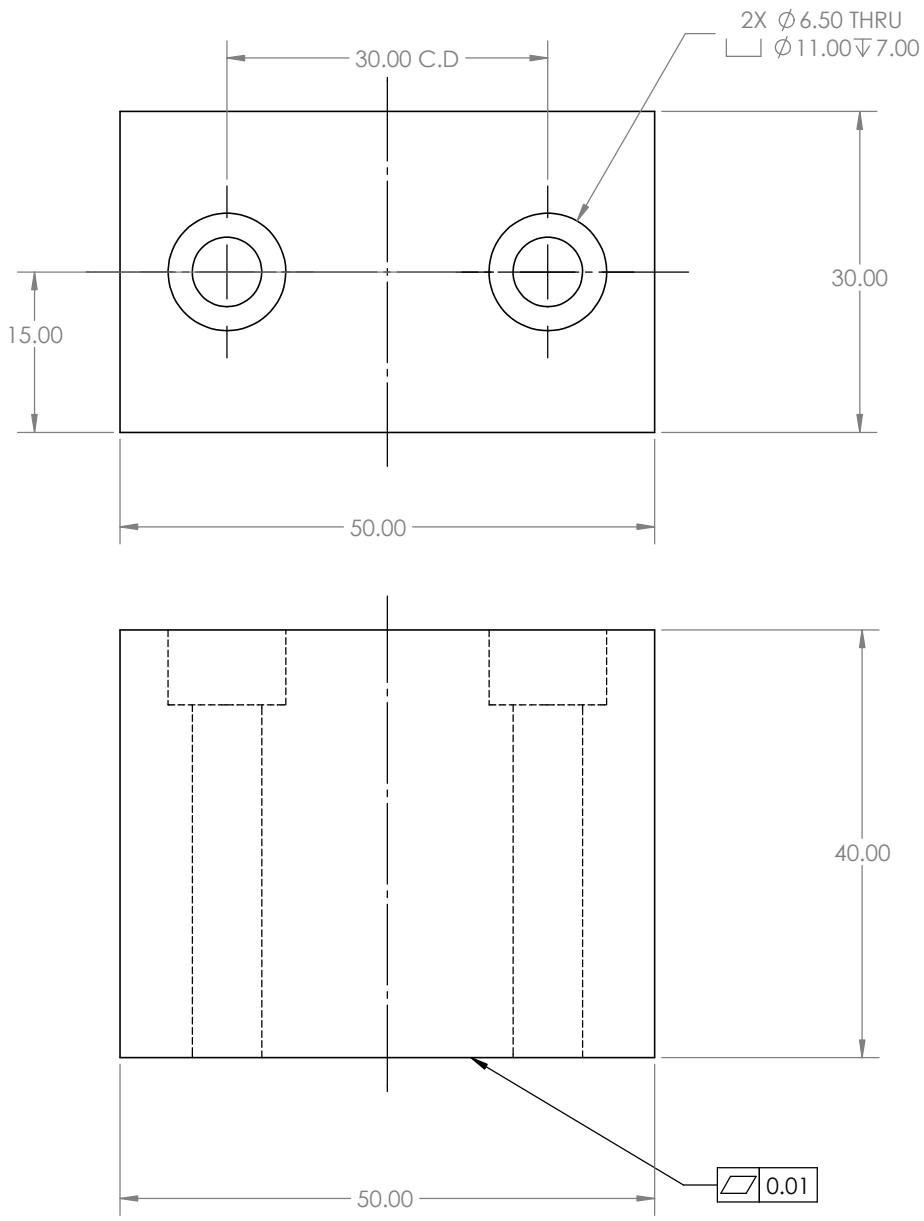
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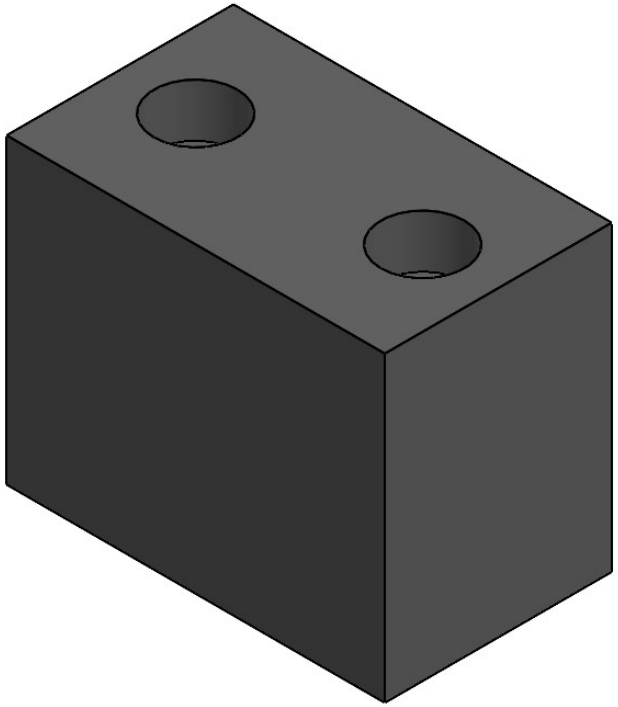
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

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MATERIAL = TEFFLON
QTY = 1 NOS

Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

TITLE
MCU SUB ASSEM STATION

Ref Drawing No.

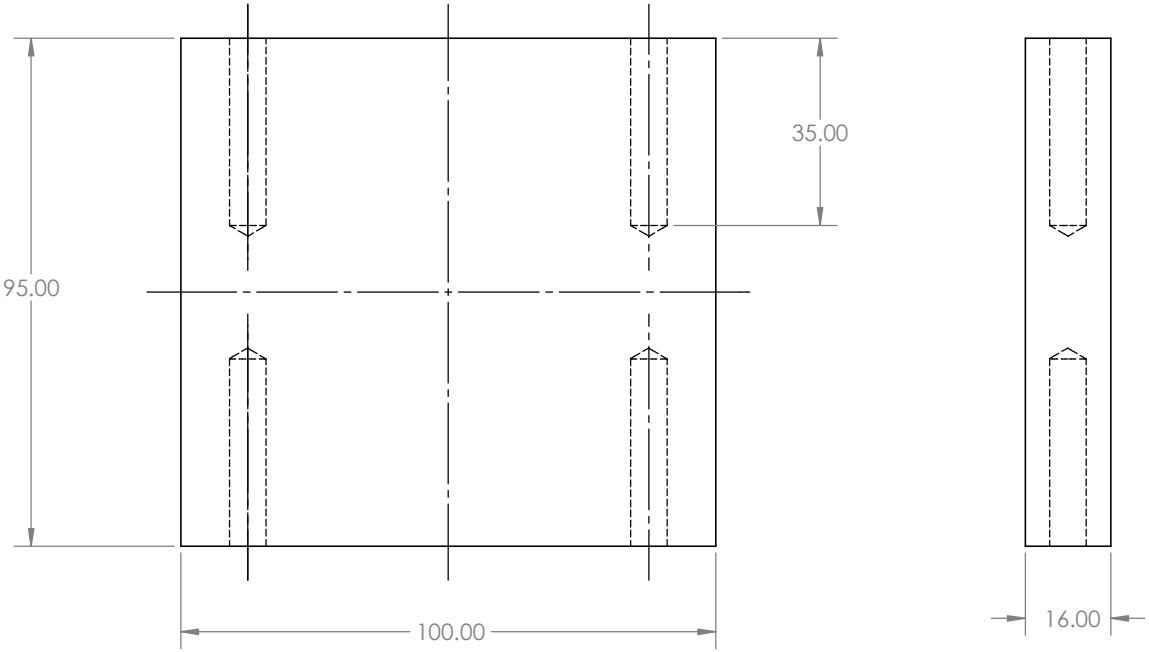
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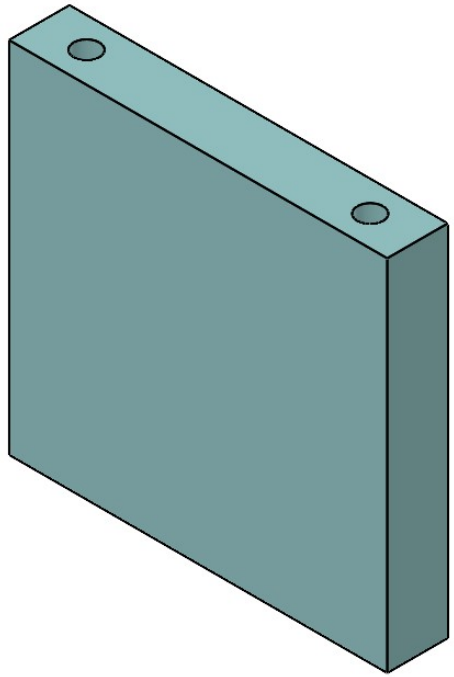
Key Characteristics (S) Safety Characteristics Critical Characteristics Important Characteristics

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MATERIAL = MS
QTY = 2 NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE MCU SUB ASSEM STATION		Ref Drawing No.	
		Drawing No.	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

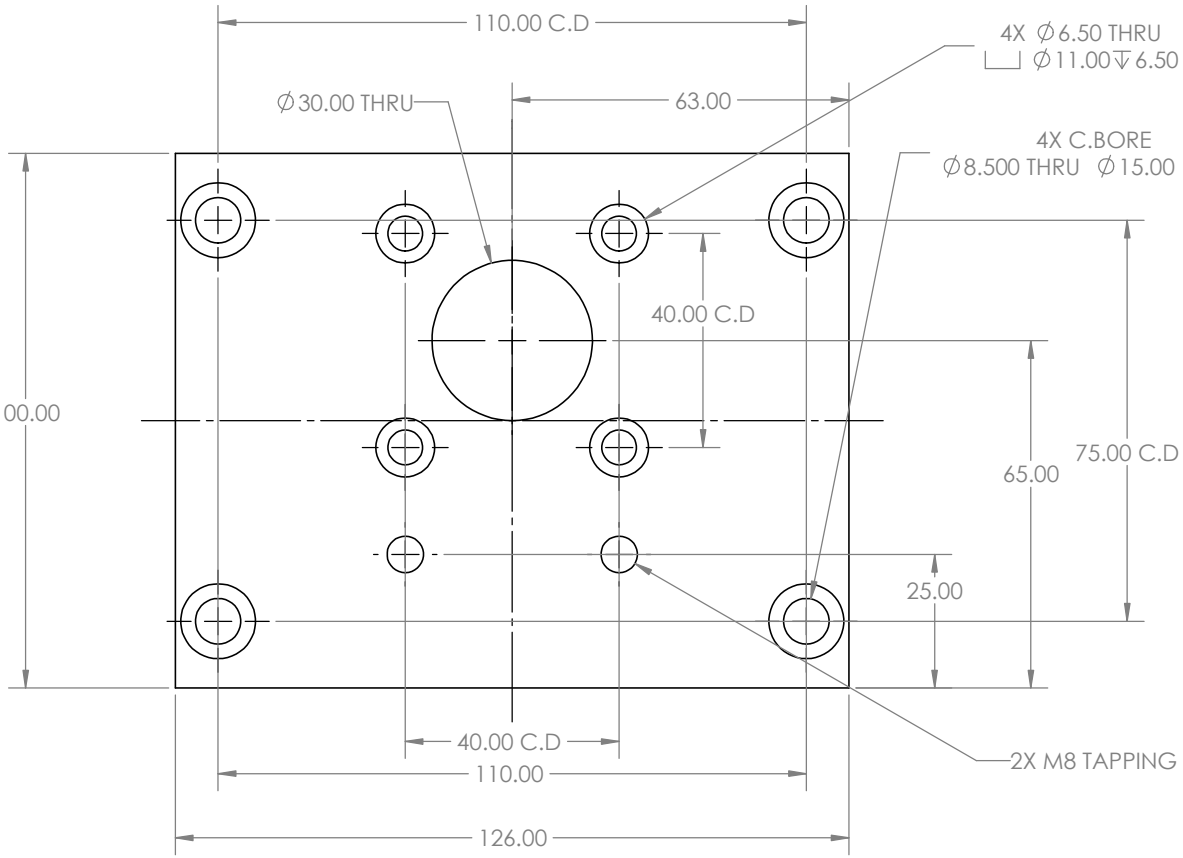
Rev

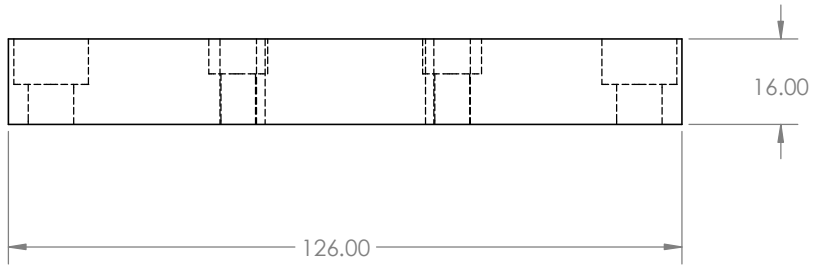
Key Characteristics | Safety Characteristics | Critical Characteristics | Important Characteristics

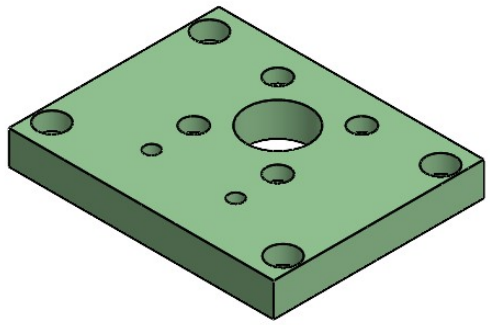
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MATERIAL = MS
QTY = 1NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Rev

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

TITLE
MCU SUB ASSEM STATION

Ref Drawing No.

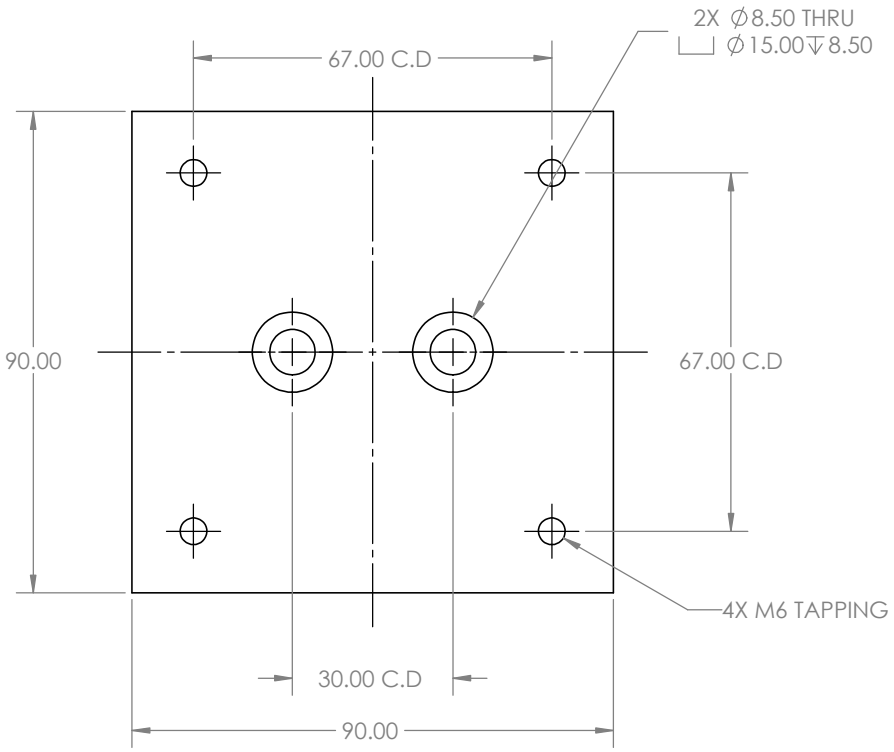
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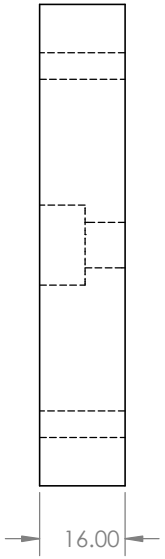
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

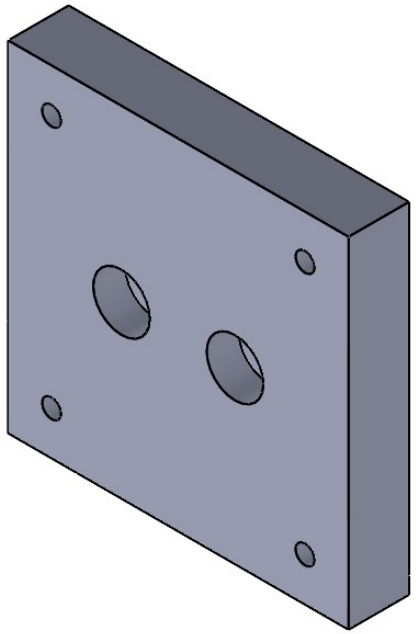
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MATERIAL = MS
QTY = 4 NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Rev

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

TITLE
MCU SUB STATION
ASSEM

Ref Drawing No.

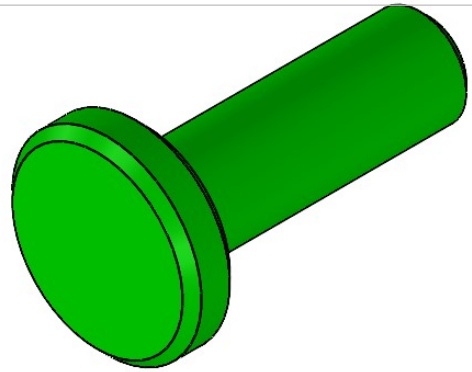
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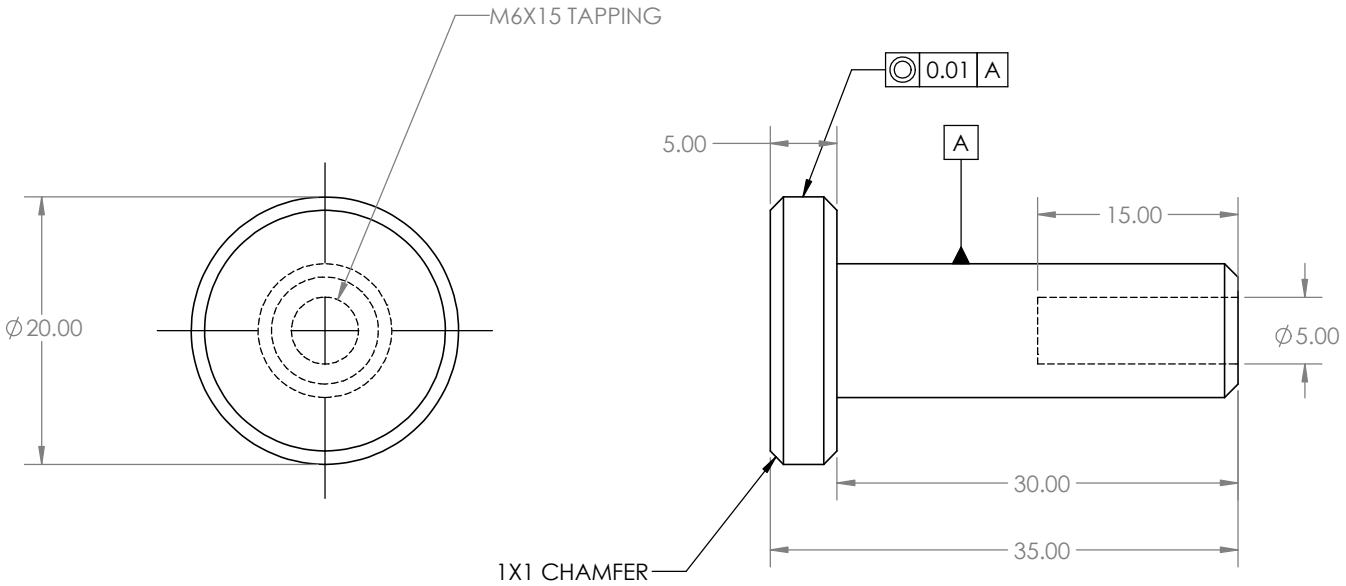
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

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MATERIAL = EN8
QTY = 2 NOS

Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE MCU SUB ASSEMSTATION		Ref Drawing No.	
		Drawing No.	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

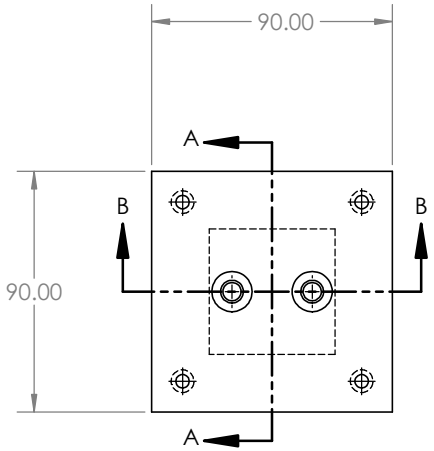
Rev

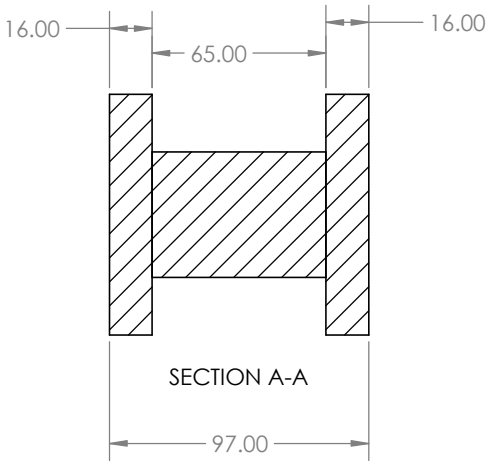
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

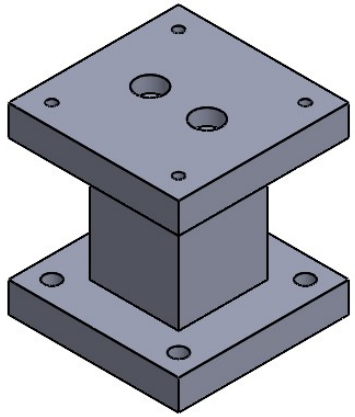
PROWESS AUTOTECH PVT. LTD. OUTSOURCE YOUR PAIN

DO NOT SCALE THE DRG.
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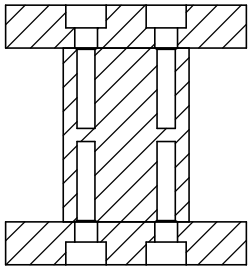
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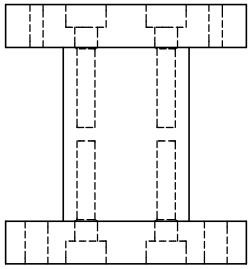




TEFFLON MTG ASSEMBLY VIEW
QTY 4 NOS



SECTION B-B



Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
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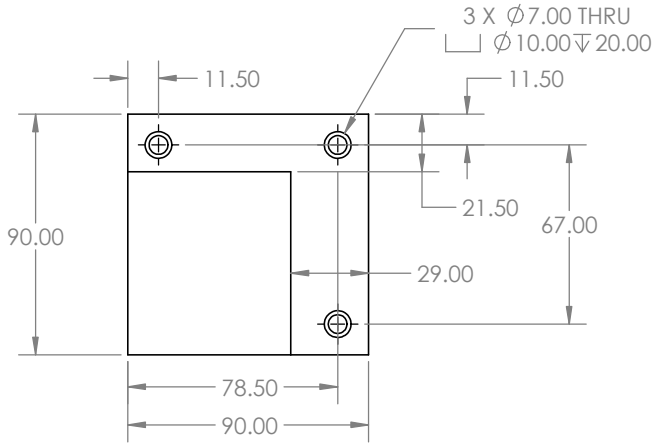
Finish		Sign.		**Prowess Autotech Pvt. Ltd.**
Drawn	Christopher		Projection	
Checked	Chanchal Pandey			MEASURE WITHOUT TOLERANCE INDICATION : REFERENCE ISO 2768-1 TOLERANCE CLASS:M GENERAL GEOMETRIC TOLERANCE REFERENCE ISO 2768-2 TOLERANCE CLASS: L
Approved	Om Prakash		Appr. Date	
TYPE		Material		
TITLE PART NAME			Ref Drawing No.	
			Drawing No.	

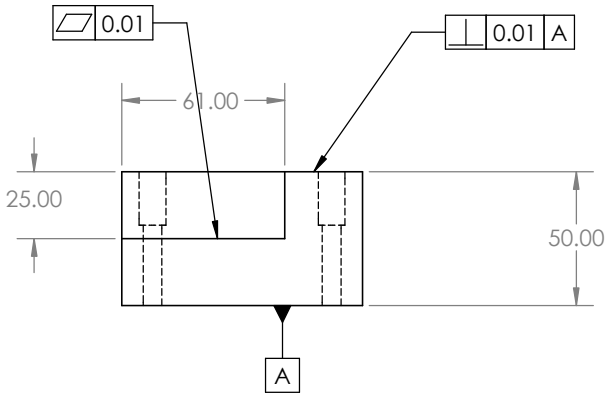
Key Characteristics | S Safety Characteristics | Critical Characteristics | Important Characteristics

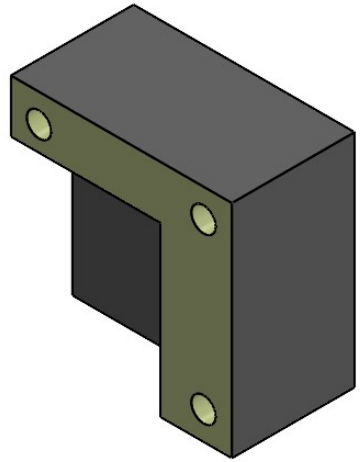
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MATERIAL = TEFFLON
QTY = 2 NOS

Rev						
Scale	Drawing No.	Description	Case Depth	Weight	Qty.	Assy Drg. No.
1:1						

Finish		Sign.	
Drawn	Christopher		Projection
Checked	Chanchal Pandey		
Approved	Om Prakash		Appr. Date
TYPE		Material	
TITLE MCU SUB ASSEMBLY STATION		Ref Drawing No.	
		Drawing No.	

Prowess Autotech Pvt. Ltd.

MEASURE WITHOUT TOLERANCE INDICATION :
REFERENCE ISO 2768-1 TOLERANCE CLASS:M
GENERAL GEOMETRIC TOLERANCE
REFERENCE ISO 2768-2 TOLERANCE CLASS: L

Key Characteristics (S) Safety Characteristics Critical Characteristics Important Characteristics

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ELCTRICAL BOM & MECHANICAL				
SR.NO	NAME	PART NO	MAKE	QTY
1	PLC	FX5U - 32M	MITSUBISHI	1
2	HMI	GS2197-WTBD-N	MITSUBISHI	1
3	POWER SUPPLY	PS-EE-2G/1AC/24DC	PHOENIX	1
4	EMERGENCY SWITCH	- - - - - - - - -	OTHER	3
5	ETHERNET SWITCH	FL 1005N	PHOENIX	1
6	ISOLATOR SWITCH	LB116	SALZER	1
7	MCB 6A(DC)	C60H-DC C 6A	SCHNEIDER	1
8	MCB 6A	A9N2P06CGN	SCHNEIDER	1
9	DOUBLE TB	KUDD4	CLMEX	20
10	TERMINAL BLOCK	CTS2.5UN	CONNECT WELL	5
11	FAN 220V	- - - - - - - - -	PHOENIX CONTACT	1
12	LED 5W	SLU5W	PHILIPS	1
13	2 POS. SELECTOR SWITCH	- - - - - - - - -	OTHER	1
14	YELLOW PB	- - - - - - - - -	TEKNIC	1
15	GREEN PB	- - - - - - - - -	TEKNIC	1
16	RED PB	- - - - - - - - -	TEKNIC	10
17	NC BLOCK	- - - - - - - - -	TEKNIC	3
18	NO BLOCK	- - - - - - - - -	TEKNIC	3
19	TOWER LIGHT	LSLO301RAGAB	LEVZE ELECTRONIC	1
20	READ SWITCH	- - - - - - - - -	SMC	4
21	FILTER	FPAV01-130*130	FTC	1
22	HOOTER LED	- - - - - - - - -	- - - - - - - - -	1
23	RELAY TB	- - - - - - - - -	PHOENIX CONTACT	8
24	1 METER LED	- - - - - - - - -	CRAFTECH	1
25	SWITCH BOARD	- - - - - - - - -	- - - - - - - - -	1
26	PNEUMATIC CYLINDER	CDQ2A40-50Z	SMC	1

PROWESS
AUTOTECH PVT. LTD.
OUTSOURCE YOUR PAIN

WARRANTY & SERVICE INFORMATION

PROJECT NO.	
DOC. NO.	
REV. NO.	00
DATE	

DETAILS

MACHINE	MANUFACTURER	CUSTOMER
MCU SUB ASSEMBLY STATION	PROWESS AUTOTECH PVT LTD.	TATA MOTORS LTD.

1 WARRANTY



- Warranty covers manufacturing defects only.
- Damage due to improper installation, misuse, unauthorized modification, or poor maintenance is excluded.
- Use only genuine spare parts.
- Warranty starts from the date of commissioning unless otherwise agreed.

2 SERVICE SUPPORT



Technical assistance, commissioning support, troubleshooting and spare parts are available from Prowess Autotech.

3 CONTACT DETAILS



Company: PROWESS AUTOTECH PVT LTD.

Address: Ground Floor, Plot No. 57, Sector IDC,
SIDCUL Industrial Area Road, IIE Pantnagar,
Rudrapur, Udham Singh Nagar, Uttarakhand - 263153

Phone: +91 9917474110

Email: prowessautotech@gmail.com

4 SERVICE REQUEST INFORMATION



DETAILS	INFORMATION
Machine No.	
Project No.	
Installation Date	
Issue Description	
Contact Person	
Remarks	

PROWESS
AUTOTECH
OUTSOURCE YOUR PAIN

DOCUMENT REVISION HISTORY

REVISION	DATE	DESCRIPTION	APPROVED BY
Rev.00	10-10-2026	Initial Release	PROWESS AUTOTECH PVT LTD

NOTES

- This manual contains important information for the safe operation and maintenance of the **MCU SUB ASSEMBLY STATION**
- Read this manual carefully before installation, operation and maintenance.
- Use only genuine spare parts for replacement.
- Follow all safety instructions and maintenance schedules for reliable performance.
- For any service support or technical assistance, contact Prowess Autotech.

SERVICE RECORD

DATE	COMPLAINT / DESCRIPTION	ACTION TAKEN	ENGINEER NAME	SIGNATURE

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PROWESS AUTOTECH PVT LTD
LOCATION - RUDRAPUR,UTTARAKHAND
PIN CODE : 263153

+91 9917474110

prowessautotech@gmail.com